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Registration number: FA22-BSE –104

Q1

Password validator

```
package model;

/**
 * @author user
 */

public class PasswordValidator {

    public boolean isValid(String password) {

        if (password == null) {
            return false;
        }

        return password.length() >= 8;
    }

}
```

Password validator test

Positive

```
package test;

/**
 *
 * @author user
 */

import model.PasswordValidator;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class PasswordValidatorTest {

    // Positive Test
    @Test
    void shouldAcceptValidPassword() {

        PasswordValidator validator =
            new PasswordValidator();

        assertTrue(
            validator.isValid("Password123")
        );
    }

    // Negative Test
    @Test
```

Negative

```

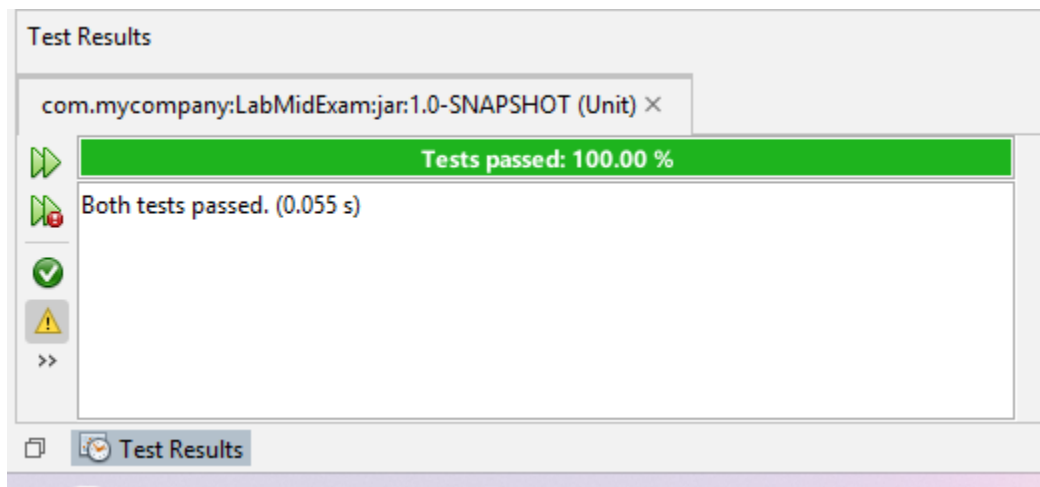
    }
    // Negative Test
    @Test
    void shouldRejectShortPassword() {

        PasswordValidator validator =
            new PasswordValidator();

        assertFalse(
            validator.isValid("abc")
        );
    }
}

```

Output

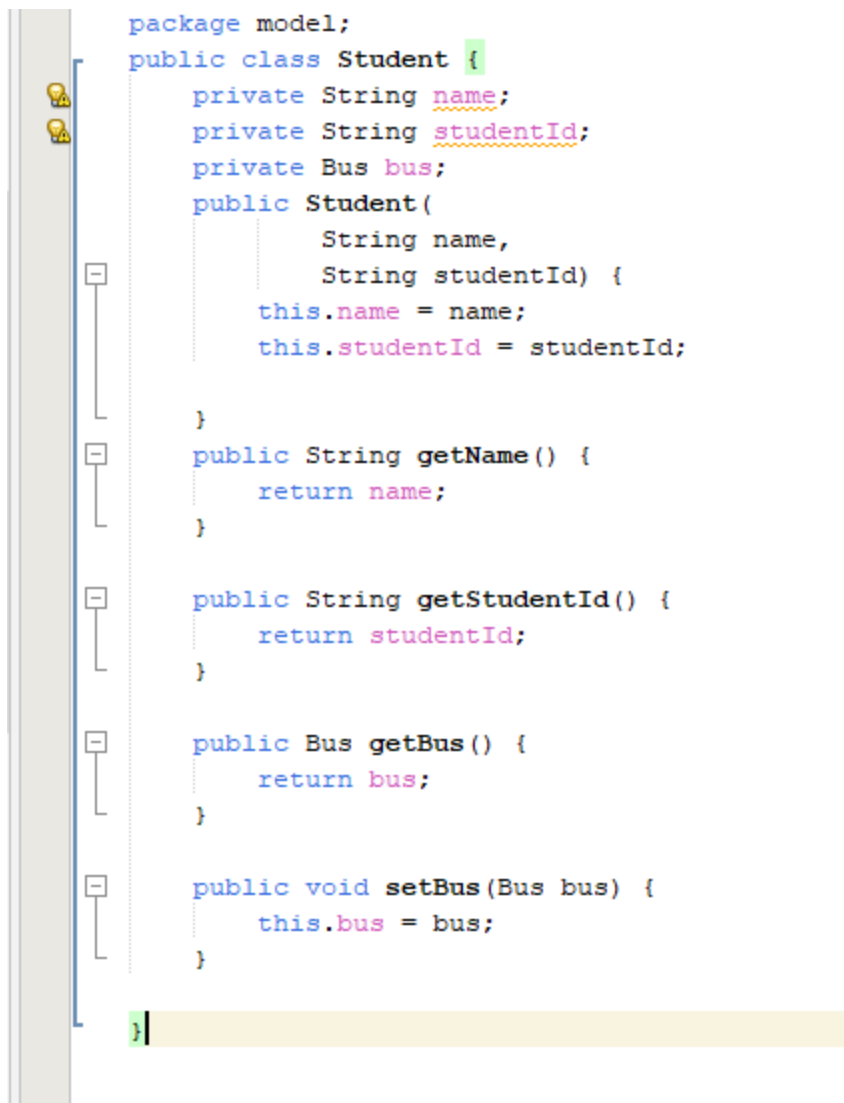


Q2

Case study : school college pick and drop management system

Use case: assign bus to students

Student



The image shows a snippet of Java code in an IDE. On the left, there is a vertical toolbar with icons for a lightbulb, a magnifying glass, and a minus sign. The code is as follows:

```
package model;

public class Student {
    private String name;
    private String studentId;
    private Bus bus;
    public Student(
        String name,
        String studentId) {
        this.name = name;
        this.studentId = studentId;
    }

    public String getName() {
        return name;
    }

    public String getStudentId() {
        return studentId;
    }

    public Bus getBus() {
        return bus;
    }

    public void setBus(Bus bus) {
        this.bus = bus;
    }
}
```

The code defines a `Student` class in the `model` package. It has three private attributes: `String name`, `String studentId`, and `Bus bus`. The class includes a constructor that takes `String name` and `String studentId` as parameters and initializes the `name` and `studentId` attributes. It also has four public methods: `getName()` which returns the `name` attribute, `getStudentId()` which returns the `studentId` attribute, `getBus()` which returns the `bus` attribute, and `setBus(Bus bus)` which sets the `bus` attribute to the provided `bus` object.

Bus

```
package model;

public class Bus {

    private String busNo;
    private String route;

    public Bus (
        String busNo,
        String route) {

        this.busNo = busNo;
        this.route = route;
    }

    public String getBusNo() {
        return busNo;
    }

    public String getRoute() {
        return route;
    }
}
```

Transport manager

```
package model;

/**
 *
 * @author user
 */

public class TransportManager {

    public void assignBus(
        Student student,
        Bus bus) {

        student.setBus(bus);

    }

}
```

Transport manager test

```

package test;
import model.Bus;
import model.Student;
import model.TransportManager;
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;
public class TransportManagerTest {
    @Test
    void shouldAssignBusToStudent() {
        TransportManager manager =
            new TransportManager();
        Student student =
            new Student(
                "Ali",
                "S001"
            );
        Bus bus =
            new Bus(
                "BUS-101",
                "Route A"
            );
        manager.assignBus(
            student,
            bus
        );
        assertEquals(
            bus,
            student.getBus()
        );
    }
}

```

Output

Test Results

com.mycompany:LabMidExam:jar:1.0-SNAPSHOT (Unit) ×


Tests passed: 100.00 %


 The test passed. (0.055 s)



